Michele Garibbo

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Education

- **PhD in Neuroscience and Machine Learning**, Wellcome Trust funded, Faculty of Engineering Mathematics, **University of Bristol**, Sep 2019 - Jan 2024, UK.
    - Focused on deep reinforcement learning, combining applications to human motor learning with methodological advances in value-based estimation.
  - Oct 2023 - Jan 2024, **Visiting PhD** experience at the **University of Oxford.**

- **Contract education in Data Science and Knowledge Engineering** (average grade achieved: 8.4/10), **Maastricht University**, Oct 2017 - Jan 2019, Netherlands.

- **MRes Cognitive Neuroscience** (Distinction, **Dean's list award -** top 5% of the Brain faculty), **University College London (UCL)**, Sep 2016 – Sep 2017, UK.

- **BSc (Hons) Psychology** (First-class, **British Psychology Association award -** top psychology graduating student), **Bath Spa University**, Sep 2013 – June 2016, UK.

Relevant experience

- **Jan 2025 - Present, Centre for Genomic Regulation, Spain, Senior Postdoctoral Associate,** Protein Language Models & Generative Protein Design (Ferruz group)

    - Led design and execution of large-scale pre-training of protein language models (ProtGPT3 project, up to 10B parameters), across multi-node, multi-GPU systems (40+ GPUs) using PyTorch Distributed, DeepSpeed, and Hugging Face Accelerate.
    - Led post-training (SFT, RLHF) of large protein and genomic (e.g., evo2) language models for target bio properties.
    - Developed large-scale evaluation pipelines for protein language model generations, integrating structure prediction (AlphaFold, ESMFold, Boltz), evolutionary consistency checks (HMM profiles, MSA-based metrics), and sequence-level scoring (ESM, ProteinMPNN) to assess structural validity and design quality at scale.

- **Apri- Dec 2024, University of Oxford, UK, Visiting Postdoctoral Research Associate,** Neuroscience and Machine Learning, under Rui Ponte Costa (link).

- **2021-2022, University of Bristol, UK, Paid Teaching assistant positions**, Engineering Mathematics 1, Computational neuroscience.

- **Feb - April 2019, Maastricht University, Maastricht, Netherlands, paid student assistantship,** Deep Reinforcement Learning and human vision**.**

- **Feb - April 2016, Annex Clinical, New York, USA, Assistant Data Analyst.** Analysis of a large clinical dataset to investigate the relation between anhedonia and depression in relation to clinical trial drop-out *(*link*)*.

Most Relevant Publications

- (Submitted) **Garibbo, M.**, Boxo, G., Middendorf, L., Stocco, F., & Ferruz, N., (2026). ProtGPT3: an Open-source family of Promptable and Aligned Protein Language Models. *NeurIPS*. (pdf)
- Stocco*, F., **Garibbo*, M.**, & Ferruz, N. (2026). Steering generative models for protein design: Aligning and conditioning strategies. Current Opinion in Structural Biology, *98*, 103250.
- Moberg, S., **Garibbo*, M.**, Mazo*, C., Gilad, A., Schmitz, D., Costa, R. P., ... & Takahashi, N. (2026). Distinct roles of cortical layer 5 subtypes in associative learning. Nature Communications, *17*(1), 2648.
- **Garibbo, M.**, Robeyns, M., & Aitchison, L. (2024). Taylor TD-learning. *Advances in Neural Information Processing Systems (NeurIPS), 37*.

**equal contribution*

Contributed Talks

- **Garibbo, M.**, Boxo, G., Middendorf, L., Stocco, F., & Ferruz, N. (2026) ProtGPT3: an Open-source family of Promptable and Aligned Protein Language Models. Eric and Wendy Schmidt Symposium on Biomedical Science and AI, Broad Institute of MIT & Harvard, USA.
- **Garibbo, M.**, Ludwig, C., Lepora, N., & Aitchison, L. (2021). *What can deep reinforcement learning tell us about human motor learning and vice-versa ?.* Neuromatch Conference, December, 2021, Online.
- **Garibbo M.** and Wierdak E. (2014). *The effect of local environmental context changes on intentional and unintentional memory retrieval*. In: British Psychology Association South West Undergraduate Conference, March 2016, University of the West of England, Bristol.

Selected Posters

- **Garibbo, M.**, Robeyns, M., & Aitchison, L. (2023). *Taylor TD-learning*. NeurIPS, USA.
- Zanzi, M., **Garibbo, M.**, Tavano, A., & Saponati, M. (2022). *RNN reconstruction of mouse latent neural dynamics*. Neuromatch Conference, Online.
- **Garibbo, M**., Ludwig, C., Lepora, N., & Aitchison, L. (2022). *What deep reinforcement learning tells us about human motor learning and vice-versa. CSHL From Neuroscience to Artificially Intelligent conference*, New York.

Programming and Technical Skills

Deep Learning frameworks: PyTorch (extensive), Hugging Face (extensive) and TensorFlow.

Distributed Training: PyTorch Distributed, DeepSpeed, Hugging Face Accelerate (multi-node, multi-GPU training up to 40+ GPUs).

Programming Language: Python (extensive), Java (basic) and Matlab.

Infrastructure & Tools:

- Linux, OS.
- Docker, Singularity, Git, Vim, Latex, high performance computing (SLURM).

Awarded Fundings

i. MINECO Colaboración Publico-Privada for a Deep Learning Scientist in protein design, 2025.
ii. Wellcome Trust 8-month Transition Fund for a Postdoctoral position, 2023.
iii. Wellcome Trust 4-year PhD Scholarship, 2019.
iv. MN5 EuroHPC Development Access Call: 14000 GPU hours, 11/2025.
v. MN5 Red Española de Supercomputación fast-track low priority GPU access, 01/2026.